

Polar Factorization and Monotone Rearrangement of Vector-Valued Functions

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ABSTRACT

We establish a fundamental polar factorization theorem for vector-valued maps, generalizing the classical polar decomposition of matrices to measure-preserving transformations. Given a probability measure μ and a cost function $c(x, y) = 1/2 |x - y|^2$, the optimal transport map T pushing μ to ν is the gradient of a convex potential: $T = \text{grad } \Phi$. In one dimension ($d = 1$), the optimal transport map is uniquely characterized as the monotone non-decreasing rearrangement. When solved via entropy-regularized Sinkhorn-Knopp flow, sorting emerges as the continuous thermodynamic relaxation towards the Birkhoff polytope vertex.

1. Monge-Kantorovich Formulation of 1D Sorting

Let $\mu = 1/n \sum_{i=1}^n \delta_{x_i}$ be an empirical probability measure on $\mathbb{R}$ representing unsorted values.

Let $\nu = 1/n \sum_{j=1}^n \delta_{y_j}$ be a reference measure representing ordered ranks $y_1 < \dots < y_n$.

The Kantorovich optimal transport problem seeks a transport coupling P in $\mathbb{R}_+^{n \times n}$ solving:

$$\min_{P \in U} = \sum_{i,j} P_{ij} C_{ij}$$

subject to: $P \mathbf{1} = 1/n \mathbf{1}$, $P^T \mathbf{1} = 1/n \mathbf{1}$, where U is the Birkhoff polytope of doubly stochastic matrices.

Brenier's Polar Factorization Theorem: If $c(x, y) = 1/2 |x - y|^2$, the optimal transport map T exists, is unique, and is the gradient of a convex function: $T = \text{grad } \Phi$.

In 1D, convex functions have non-decreasing derivatives (Φ' is monotone non-decreasing). Therefore, Brenier's theorem guarantees that optimal transport in 1D is strictly the monotone sorting rearrangement!

2. Entropic Regularization & Sinkhorn Flow

To compute optimal transport continuously, Cuturi (2013) introduces entropic regularization:

$$\min_{\mathbf{P} \in \mathcal{U}} -\epsilon \mathbf{H}(\mathbf{P})$$

where $\mathbf{H}(\mathbf{P}) = -\sum_{i,j} P_{ij} (\ln P_{ij} - 1)$ is the Shannon entropy.

The unique optimal solution is $\mathbf{P}_\epsilon = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v})$ with Gibbs kernel $\mathbf{K} = \exp(-\mathbf{C} / \epsilon)$.

Sinkhorn-Knopp Balancing: Alternating projections $\mathbf{u} \leftarrow (1/n) / (\mathbf{K} \mathbf{v})$ and $\mathbf{v} \leftarrow (1/n) / (\mathbf{K}^T \mathbf{u})$ converge exponentially fast.

Properties: (1) Infinite Differentiability (C^∞), (2) Asymptotic Exactness: as $\epsilon \rightarrow 0+$, $\mathbf{P}_\epsilon$ converges to the exact discrete permutation matrix, and (3) Barycentric Soft Sorting: $\mathbf{s} = n \mathbf{P}_\epsilon^T \mathbf{x}$ is a differentiable relaxation of $\text{sort}(\mathbf{x})$.